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SECURITY BAG AND METHOD OF USE

Abstract

A security bag has a storage pouch that is shaped to receive and securely contain a smart phone. Front and rear panels of the storage pouch are constructed of an opaque material to prevent operation of cameras of the smart phone. A central portion of the front panel has a control opening that has a height and a width that are adapted to allow access to at least part of the touch screen of the smart phone, without uncovering the camera. A locking mechanism locks the access opening closed and prevents removal of the smart phone from the storage pouch. In an unlocked position the locking mechanism may be physically detached from the storage pouch.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] This invention relates generally to security bags, and more particularly to a security bag that securely stores a smart phone, preventing any photography using the smart phone, but still enabling a user to have partial access to the smart phone for texting and other uses.

Description of Related Art

[0002] Dugoni, U.S. Pat. No. 9,819,788, teaches a security pouch that is adapted for containing smart phones during events such as concerts, comedy shows, etc. The security pouch has a durable neoprene construction and an integral lock mechanism that enables attendees to maintain physical control over their phones, while completely blocking the user from using or accessing the phone in any way until the event ends, or until the user physically leaves a designated physical space of the event.

[0003] A first disadvantage of this approach is that there are occasions where a user might prefer or require at least limited use of their phone. Complete denial of any phone access is not acceptable in many circumstances, and there is a need in the art for a security device that denies access to cameras but allows some access (e.g., texting, calls, etc.) while the phone is in the pouch.

[0004] A second disadvantage of this approach is that the Dugoni pouch is constructed of a rugged material such as neoprene, and the lock is integrated with the pouch, so that the total cost of the unit is relatively high, and it is intended for long term use. It would not be practical to use this form of pouch only a single time, and either dispose of the pouch, or allow the user to keep it as a commemorative item. But there is a need in the field for a pouch that may economically be used only a single time, and then either thrown away, or kept as a collectable item. Not only does this remove problems with spreading contagions and similar problems arising from multiple commercial uses of the pouch, it also adds value in the form of a collectable item that may be greatly valued by the users who keep the pouch after use. The locking mechanism of the current invention may be completely removed from the pouch, so that the lock may be re-used while the pouch itself is released and not re-used.

[0005] The present invention fulfills these needs and provides further advantages as described in the following summary.

SUMMARY OF THE INVENTION

[0006] The present invention teaches certain benefits in construction and use which give rise to the objectives described below.

[0007] The present invention provides a security bag that securely stores a smart phone for preventing use of cameras of the smart phone, but still enables a user to have operative control access to a touch screen of the smart phone. The security bag comprises a storage pouch having a front panel and a rear panel joined together to form a closed bottom end and an access opening at a top end, the storage pouch being shaped to receive and securely contain the smart phone. The front and rear panels are constructed of an opaque material to prevent operation of the cameras of the smart phone. A central portion of the front panel of the storage pouch has a control opening that has a height and a width that are adapted to cover the cameras but allow access to at least part of the touch screen of the smart phone. A transparent plastic window covers the control opening and enables operation of the smart phone via the touch screen. A locking mechanism moves between a locked position and an unlocked position. In the locked position the locking mechanism locks the access opening closed and prevents removal of the smart phone from the storage pouch, and in the unlocked position the locking mechanism may be physically detached from the storage pouch, thereby allowing removal of the smart phone.

[0008] A primary objective of the present invention is to provide a security bag having advantages not taught by the prior art.

[0009] Another objective is to provide a security bag that is adapted to receive a smart phone and

lock so that the smart phone cannot be removed without the use of an unlocking device.

[0010] Another objective is to provide a security bag that includes a lock that is completely detachable from the pouch, so that the pouch may be single-use and the lock may be reused.

[0011] A further objective is to provide a security bag that is adapted to receive a smart phone, wherein a transparent window of the security bag allows limited touch screen access to the smart phone.

[0012] Other features and advantages of the present invention will become apparent from the following more detailed description, taken in conjunction with the accompanying drawings, which illustrate, by way of example, the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] The accompanying drawings illustrate the present invention.

[0014] FIG. 1 is an exploded perspective view of a security bag according to one embodiment of the present invention, illustrating a storage pouch and a locking mechanism of the security bag.

[0015] FIG. 2 is a rear elevational view illustrating a smart phone being inserted into the storage pouch.

[0016] FIG. 3 is a perspective view of the security bag illustrating the smart phone positioned within the storage pouch and the locking mechanism in place to lock the storage pouch closed.

[0017] FIG. 4 is a sectional view taken along line 4-4 in FIG. 3, illustrating the locking mechanism in a locked position on the storage pouch.

[0018] FIG. 5 is a sectional view similar to FIG. 4, illustrating the locking mechanism being unlocked with a magnetic key.

DETAILED DESCRIPTION OF THE INVENTION

[0019] The above-described drawing figures illustrate the invention, a security bag that securely stores a smart phone for preventing use of cameras of the smart phone, but still enables a user to have operative control access to a touch screen of the smart phone. Such a security bag may be used in a wide variety of uses, including concerts and similar shows, in schools, in various legal and government locations, and any other location or situation where the camera function of the smart phone may need to be limited.

[0020] FIG. 1 is an exploded perspective view of a security bag **10** according to one embodiment of the present invention, illustrating a storage pouch **20** and a locking mechanism **40** of the security bag **10**. FIG. 2 is a rear elevational view illustrating a smart phone **12** being inserted into the storage pouch **20**. FIG. 3 is a perspective view of the security bag **10** illustrating the smart phone **12** positioned within the storage pouch **20** and the locking mechanism **40** in place to lock the storage pouch **20** closed.

[0021] As shown in FIGS. 1-3, the storage pouch **20** has a front panel **22a** and a rear panel **22b** joined together to form a closed bottom end **24** and an access opening **26** at a top end **28**, the storage pouch **20** being shaped to receive and securely contain the smart phone **12**. In the current embodiment, the front panel **22a** and the rear panel **22b** are a single piece of material that is folded along a bottom edge, and side edges are fastened together (e.g., sewn, welded, etc.). However, in other embodiments, other methods of forming the pouch **20** may be used.

[0022] The security bag **10** may further include a closure **30** for covering the access opening **26**. In this embodiment, the closure **30** is in the form of a flap that extends from the front panel **22a** and can fold over the access opening **26**, and then contact the rear panel **22b** for receiving the locking mechanism **40**. As shown in FIG. 2, the flap **30** may have a closure hole **32** that aligns with front and rear holes **34** of the front and rear panels **22** when the storage pouch **20** is in a closed position (FIGS. 1 and 3), wherein all three holes **32** and **34** receive the locking mechanism **40**, discussed in

greater detail below. In other embodiments, other forms of closures known in the art may be used. The three holes **32** and **34** may be formed in a lateral center of the security bag **10** adjacent the top end **28**, but may also be formed off-center in other embodiments. Alternatively, the closure **30** may be any other form of closure, e.g., a zip-seal closure, button, etc., or alternatively the security bag **10** may be closed only by the locking mechanism **40**. The locking mechanism **40** is further illustrated in FIGS. **4-5** and discussed in detail below.

[0023] In some embodiments, the front panel **22a** and/or the rear panel **22b** may be printed with commemorative images **36** that are collectable in nature. FIG. **2** shows one example of commemorative images being in the form of an awards ceremony design. Obviously, any commemorative images may be applied, for a wide range of potential events, e.g., a sporting event, personal party, comedy club, etc. In other embodiments, only a logo or identifying numbers/lettering may be added, or the panels may be left blank. The images **36** may be printed, embossed, or formed in any other printing or marking method known in the art. Such images **36** are not used in the prior art because the prior art pouches are not disposable, but must be reused for a significant period of time to justify their expense. The present invention may be custom printed for each event in which it is used, because they are only used once, and only the removable lock is reused.

[0024] As shown in FIG. **3**, the front and rear panels **22** are constructed of an opaque material to prevent operation of the cameras **14** of the smart phone **12**, while a central portion of the front panel **22a** of the storage pouch **20** has a control opening **35** with a height H and a width W that are adapted to allow access to at least part of the touch screen of the smart phone **12**, while keeping the cameras **14** covered. A transparent plastic window **38** may cover the control opening **35** and enables operation of the smart phone **12** via the touch screen, while the opaque material of the front and rear panels **22** cover the cameras **14** of the smart phone **12**. In some embodiments, the height H of the control opening **35** may be between 3-5 inches, and the width W may be between 2-3 inches. However, in alternative implementations, the security bag **10** may be adapted to instead contain a larger device such as a tablet computer, and the height and width would then be adjusted to expose the desired portions of the given touch screen. Furthermore, the security bag **10** may also be adapted to contain a smaller device such as a “mini” version of a smart phone, a large smart watch, or similar. In use, various different-sized security bags may be provided at once, to accommodate the different devices of event attendees. In this embodiment, the part of the touch screen that is visible includes the full touch screen except for the front camera region (not shown). In different embodiments, less of the touch screen may be exposed, depending on the desires of the operators of the locking mechanism **40**.

[0025] In this embodiment, the front and rear panels **22** are constructed of paper or plastic, and are disposable in nature, so that a user of the security bag **10** may keep the storage pouch **20** after it has been used, although the more expensive locking mechanism **40** will be reused many times before it is replaced. However, in some embodiments, the security bag **10** and locking mechanism **40** may both be re-usable. Other opaque materials may be used for the front and rear panels **22**, e.g., neoprene, if a non-disposable version is desired.

[0026] FIG. **4** is a sectional view taken along line **4-4** in FIG. **3**, illustrating the locking mechanism **40** in a locked position on the storage pouch **20**. FIG. **5** is a sectional view similar to FIG. **4**, illustrating the locking mechanism **40** being unlocked with a magnetic key **70**. The locking mechanism **40** moves between a locked position (shown in FIG. **4**) and an unlocked position (shown in FIG. **5**). As shown in FIG. **4**, in the locked position the locking mechanism **40** locks the access opening **26** closed and prevents removal of the smart phone **12** from the storage pouch **20**.

[0027] As shown in FIGS. **4-5**, in this embodiment, the locking mechanism **40** includes a front plate **42** connected to a rear plate **44** with a hinge **46**, the locking mechanism **40** further including a locking tab **48** extending inwardly from the rear plate **44**, and a locking pin **60** of the front plate **42** that lockingly engages the locking tab **48**. The locking pin **60** is biased towards a locked position

by a spring **50**, but movable to an unlocked position to release the locking tab **48** and open the locking mechanism **40**. In various embodiments, other biasing mechanisms may be used, e.g., other forms of springs, pads/cushions, resilient tabs, etc., or any other form of resilient element.

[0028] As illustrated, the locking mechanism **40** may include a pin housing structure **52** that extends beneath the locking tab **48** when the locking mechanism **40** is hinged in a closed position. The pin housing structure **52** may include a lower sheet **54** that extends inwardly from the bottom of the front plate **42**, wherein the spring **50** of the locking pin **60** is positioned on top of the lower sheet **54**. An upwardly extending sidewall **56** may extend upwardly from the lower sheet **54**, behind the rear plate **44**, so that the spring **50** is surrounded by the front plate **42**, lower sheet **54**, and upwardly extending sidewall **56**. The sidewall **56** may extend upwardly to a first inwardly extending lip **58a**, wherein a second inwardly extending lip **58b** extends from the front plate **42** opposite the first inwardly extending lip **58a** to form an upper opening **59**. As shown in FIG. **4**, in the locked position, the flap **30** of the security bag **10** and the front and rear panels **22** are positioned between the sidewall **56** and the rear plate **44**, so that the locking tab **48** extends through the closure hole **32** of the flap **30**, the rear hole **34b** of the rear panel **22b**, and the front hole **34a** of the front panel **22a**.

[0029] As shown in FIGS. **4-5**, the locking pin **60** may comprise an upper end **64** that engages with an aperture **49** in the locking tab **48**, and a lower end **66** attached to the spring **50**, wherein an annular protrusion **68** is positioned between the upper and lower ends **64** and **66**, and the annular protrusion **68** is larger in diameter than the rest of the locking pin **60**. In this manner, the upper end **64** of the pin **60** is permitted to pass through the upper opening **59** of the pin housing structure **52**, and into the aperture **49** of the locking tab **48**, while the annular protrusion **68** prevents the rest of the pin **60** from passing through the upper opening **59**.

[0030] As shown in FIG. **5**, in the unlocked position the locking mechanism **40** may be physically detached from the storage pouch **20**, thereby allowing removal of the smart phone **12**. As discussed, the magnetic key **70** may retract locking pin **60** and/or spring **50** so that the upper end **64** of the locking pin **60** releases the locking tab **48** and the locking mechanism **40** may be hinged open to release the storage pouch **20** and remove the smart phone **12** from the storage pouch **20**. While a simple magnetic key **70** is illustrated, in some embodiments, a different unlocking tool (not shown) may be used, wherein the unlocking tool may utilize elements other than magnets to retract the locking pin **60**, e.g., protrusions that frictionally engage the locking pin **60**, and/or biasing elements through the lower sheet **54** of the pin housing structure **52**, some form of suction mechanism, etc., or any other form of retracting the biasing element.

[0031] The title of the present application, and the claims presented, do not limit what may be claimed in the future, based upon and supported by the present application. Furthermore, any features shown in any of the drawings may be combined with any features from any other drawings to form an invention which may be claimed.

[0032] As used in this application, the words “a,” “an,” and “one” are defined to include one or more of the referenced item unless specifically stated otherwise. The terms “approximately” and “about” are defined to mean $\pm 10\%$, unless otherwise stated. Also, the terms “have,” “include,” “contain,” and similar terms are defined to mean “comprising” unless specifically stated otherwise. Furthermore, the terminology used in the specification provided above is hereby defined to include similar and/or equivalent terms, and/or alternative embodiments that would be considered obvious to one skilled in the art given the teachings of the present patent application. While the invention has been described with reference to at least one particular embodiment, it is to be clearly understood that the invention is not limited to these embodiments, but rather the scope of the invention is defined by claims made to the invention.

Claims

1. A security bag that securely stores a smart phone for preventing use of cameras of the smart phone, but still enables a user to have operative control access to a touch screen of the smart phone, the security bag comprising: a storage pouch having a front panel and a rear panel joined together to form a closed bottom end and an access opening at a top end, the storage pouch being shaped to receive and securely contain the smart phone, the front and rear panels being constructed of an opaque material to prevent operation of the cameras of the smart phone; a central portion of the front panel of the storage pouch having a control opening that has a height and a width that are adapted to allow access to at least part of the touch screen of the smart phone, while leaving the cameras covered; a transparent plastic window that covers the control opening and enables operation of the smart phone via the touch screen; and a locking mechanism that moves between a locked position and an unlocked position, wherein in the locked position the locking mechanism locks the access opening closed and prevents removal of the smart phone from the storage pouch, and wherein in the unlocked position the locking mechanism may be physically detached from the storage pouch, thereby allowing removal of the smart phone.
 2. The security bag of claim 1, wherein the front and rear panels are constructed of paper or plastic.
 3. The security bag of claim 1, wherein the front and rear panels are constructed of paper or plastic, and are printed with commemorative images that are collectable in nature.
 4. The security bag of claim 1, wherein the locking mechanism includes a front plate connected to a rear plate with a hinge, the locking mechanism further including a locking tab extending from the rear plate, and a locking pin of the front plate that lockingly engages the locking tab, the locking pin being biased towards a locked position by a spring, but movable to an unlocked position to release the locking tab and open the locking mechanism.
 5. The locking mechanism of claim 4, further comprising a pin housing structure that extends beneath the locking tab when the locking mechanism is hinged in a closed position, the pin housing structure comprising: a lower sheet that extends inwardly from the bottom of the front plate; an upwardly extending sidewall that extends upwardly from the lower sheet, behind the rear plate, so that the spring is surrounded by the front plate, lower sheet, and upwardly extending sidewall; a first inwardly extending lip extending inwardly from the upwardly extending sidewall, and a second inwardly extending lip extending inwardly from the front plate opposite the first inwardly extending lip to form an upper opening that allows the locking pin to pass at least partially through the upper opening to engage the locking tab.
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